Differentially Private Link Prediction in Federated Setting

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***Abstract*—Federated link prediction is difficult to deploy privately: interaction edges remain distributed across clients, yet a useful predictor must exploit graph structure and answer candidate-pair queries without rereading the private graph. A second difficulty is less visible but equally important: under edge differential privacy (DP), a structural channel can improve one target graph and harm another, so unconditional deployment can be worse than a predictor using only public node descriptors. We introduce CertFed-LP, an architecture-agnostic policy that combines an inference-closed edge-DP structural learner with a private, target-domain utility certificate. CertFed-LP activates the structural branch only when a one-sided lower certificate exceeds a registered material-gain threshold; otherwise it falls back to the public branch. We prove that no training-only selector can provide a distribution-free nontrivial no-harm guarantee, establish complete training-and-certification transcript privacy, and derive a finite-population no-harm theorem. Sufficient and necessary certification counts match in their principal dependence on effect gap, privacy budget, and record dependence. In a preregistered one-time evaluation on six social and blog networks with five seeds, the primary visible-message mechanism has composed privacy ($\epsilon = 5.664, \delta = 2 \times 10^{-6}$). It activates 15 of 30 dataset-seed cells across three datasets, observes no material false activation, and obtains mean disjoint-holdout pairwise gain 0.0854 over the public branch. An always-structural policy is harmful in 10 of 30 cells and achieves lower mean gain. The result supports certified fallback, rather than universal structural superiority, as a defensible deployment principle for federated edge-private link prediction.**

***Index Terms*—Differential privacy, federated graph learning, link prediction, private certification, privacy-utility trade-off.**

I. Introduction

Link prediction supports friend recommendation, interaction discovery, and collaborative filtering. In many deployments, however, the edges needed for training do not reside in one trusted graph: institutions or clients own disjoint interaction records and are not allowed to disclose raw neighborhoods. Federated optimization limits raw-data movement, but it does not by itself protect an edge from a server that observes model updates or released representations [1], [2]. Edge-level differential privacy (DP) supplies a record-level guarantee, but graph message passing makes both sensitivity and inference subtle. A model that consults the private graph again when answering a link query can invalidate an otherwise correct training-time accountant [3].

Prior work addresses important portions of this problem. GAP makes private graph aggregation inference-closed [3]; DPLP derives subgraph sensitivity for centralized link prediction [4]; and decentralized or federated graph learners protect links, features, representations, or local training processes under several trust models [5], [6], [7], [8]. Private recommendation further shows that edge-DP prediction is meaningful in heterogeneous and collaborative-filtering settings [9], [10], [11]. These results make a broad "first federated private link prediction" claim untenable. They also expose a remaining deployment question:

> *When a frozen edge-DP structural branch is useful only on some target graphs, can a federated service activate it without sacrificing either edge privacy or utility relative to a public-input-only fallback?*

This question cannot be answered by reporting only the accuracy of a selected private model. Public descriptors can already be predictive, DP noise scales with the released structural state, and target graphs differ in how much additional link signal their private topology contains. In our experiments, the same frozen structural mechanism produces positive pairwise gains on BlogCatalog, GitHub Social, PolBlogs, and Deezer, but negative gains on Facebook and LastFM. A rule that always enables private structure is therefore not a no-harm policy.

We propose CertFed-LP, illustrated in Figure 1. It separates the *candidate learner* from the *deployment decision*. Any admitted candidate must release an inference-closed edge-DP state R. On a disjoint set of private certification edges, CertFed-LP computes a bounded candidate-minus-public pairwise advantage, releases only a Gaussian-noised sum and count, and activates the candidate when a one-sided lower certificate exceeds a predeclared material gain γ. Otherwise it returns the public branch. All prediction scores are post-processing of public inputs, R, and the private binary decision.

Our contributions are:

- We formulate certified deployment for ordinary-graph federated link prediction under add/remove edge adjacency, complete server-transcript accounting, cross-client candidates, and inference-closed scoring.
- We prove an indistinguishable-worlds result showing that no training-only selector has a distribution-free, nontrivial no-harm guarantee. Direct target-domain certification is therefore not merely a learned dataset selector.
- We establish a finite-population no-harm theorem for an edge-keyed random certification split. We also derive a sufficient activation boundary and a central-DP testing lower bound with matching principal dependence $O(\chi/\alpha^2 + 1/(\epsilon\alpha))$ up to constants and logarithms.
- We conduct a preregistered, one-time, five-seed study on six social and blog networks. CertFed-LP safely activates in 15/30 primary cells, avoids all ten harmful always-structural cells, and reaches within 0.0057 mean pairwise

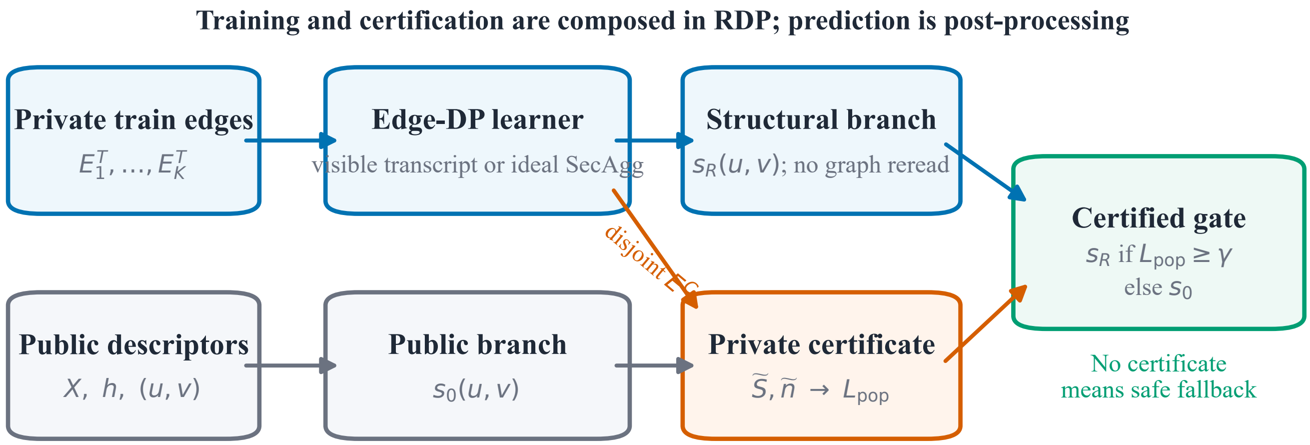

Fig. 1. CertFed-LP separates learning from deployment. The structural learner releases an inference-closed edge-DP state from training edges E^T. A disjoint certification set E^C contributes only a bounded utility sum and count. The structural branch is enabled only when the private lower certificate reaches γ; all other cases fall back to the public branch. The R5 implementation conservatively composes training and certification RDP curves.

gain of a non-deployable sign oracle.

We do not claim that the registered pairwise certificate is an AUC confidence interval, that one private learner is universally best, or that ideal secure aggregation follows from DP. These boundaries are central to the method.

II. Related Work

A. Private graph learning and inference closure

DP graph learning must control how one edge affects aggregation, optimization, and inference. GAP perturbs bounded graph aggregations and caches the released channels, so subsequent prediction does not query the private adjacency [3]. This inference-closure principle is part of our admission contract. DPLP instead studies centralized subgraph-based link prediction and bounds the dependency induced by neighborhood and path subgraphs [4]. Our contribution is not a replacement GNN architecture. It is a private policy for deciding whether any admitted inference-closed structural branch should be deployed on a target graph.

B. Decentralized and federated private graphs

Solitude collects decentralized adjacency information with local edge DP [5]. LGA-PGNN combines local perturbation, graph augmentation, and attack evaluation [6]. PDGL jointly protects decentralized neighbor lists, features, and labels using DP and cryptographic components [7]. PrivFGL addresses DP-induced client heterogeneity in federated node classification [8]. FED-PUB studies personalized learning over distributed subgraphs without the same edge-DP output contract [13], while Tang and Hu perturb federated GNN training for link prediction under a local trust model [14]. FedHGPP combines edge-DP structural preprocessing with encrypted federated heterogeneous recommendation [11]. These systems establish that distributed ownership, edge privacy, and link-oriented evaluation are not novel by themselves. We study a different output contract: ordinary interaction edges are partitioned across clients; the server-visible training and certification transcripts are accounted; and the released candidate-pair scorer cannot reread the private graph.

C. Private link prediction and recommendation

Federated knowledge-graph embedding has combined DP and link-oriented tasks under embedding-record adjacency [12]. For ordinary graphs, DPLP is the closest formal centralized predecessor [4]. PP-HGRL protects auxiliary heterogeneous relations for recommendation, while its server owns the primary user-item graph and performs centralized training [9]. CF-DPGNN applies subgraph sampling and DP-SGD to centralized collaborative filtering [10]. FedHGPP is a closer federated heterogeneous recommendation predecessor, but its objective is learner utility under a two-stage edge-DP preprocessing pipeline rather than private target-domain branch certification [11]. Our novelty is therefore not the combination of federation, edge privacy, and link-oriented prediction. CertFed-LP addresses heterogeneous *deployment utility*: it can wrap these or future learners once their privacy and inference contracts are made compatible.

D. Private hypothesis testing

The certificate is a private one-sided test over bounded utility records. Our lower-bound argument uses DP versions of Le Cam's method [15], while the finite-population analysis uses sampling-without-replacement concentration [16]. Unlike generic private model selection, CertFed-LP releases one frozen binary branch decision and explicitly includes the certification privacy cost.

TABLE I
POSITIONING BY TASK, OWNERSHIP, PRIVACY OBJECT, AND RELEASED INFERENCE. "NC" DENOTES NODE CLASSIFICATION. ENTRIES DESCRIBE THE FORMAL SCOPE OF EACH CITED WORK RATHER THAN RANKING IMPLEMENTATIONS.

Work	Ownership / task	Formal privacy object	Released inference contract
FKGE [12]	Peer knowledge graphs / KG LP	PATE-style privacy for generated embeddings	Cross-domain embeddings; not ordinary-edge adjacency
GAP [3]	Central ordinary graph / NC	Add/remove edge- and node-DP aggregation	Cached noisy channels; no adjacency reread
DPLP [4]	Central ordinary graph / LP	Edge-private neighborhood/path subgraphs	Subgraph link predictor; no federated transcript
Solitude [5]	Decentralized adjacency lists / NC	Local edge- and feature-DP collection	Server reconstructs a calibrated noisy graph
LGA-PGNN [6]	Decentralized local graphs / NC	Feature-oriented local perturbation	Class probabilities with attack evaluation
PDGL [7]	Node-held graph records / NC	Edge-LDP plus cryptographic protocols	Secure per-node label inference
PrivFGL [8]	Cross-silo subgraphs / NC	Reported node-LDP/CDP transformation	Federated classifier; no LP score contract
PP-HGRL [9]	Client auxiliary relations, server primary graph / recommendation	Perturbed auxiliary edges and hidden states	Central recommendation over server-held interactions
FedHGPP [11]	Federated heterogeneous graphs / recommendation	Two-stage pure edge-DP preprocessing	Encrypted embedding aggregation; HR/NDCG outputs
CertFed-LP (ours)	Distributed ordinary edges / LP	Add/remove role-labelled edge DP for full transcript	Inference-closed scorer plus private utility gate

III. PROBLEM AND PRIVACY MODEL

A. Federated ordinary graph

Let V be a public node universe, X registered public node descriptors, and $h : V \to [K]$ a public home-client map. Client k owns canonical undirected edges E_k under a public ownership rule; every edge is owned once. Two federated databases are adjacent when one role-labelled canonical edge is added to or removed from one client. Features, labels, node existence, and the home map are outside this edge-DP guarantee.

At the protocol level, an edge-stable pseudorandom rule can assign records to training E^T, certification E^C, or research evaluation E^Q. The deployed mechanisms never read E^Q. R5 takes the frozen role labels as part of its protected database and deliberately uses sequential composition rather than parallel-composition credit. Its historical raw-graph split was not designed to establish that adding an edge before partitioning leaves every other role unchanged; consequently, the reported guarantee is not silently upgraded to raw pre-partition graph adjacency.

B. Two inference-closed branches

The public branch is

$$s_0(u, v) = F_0(X, u, v; \omega_0), \tag{1}$$

where ω_0 is public randomness. A DP trainer releases state

$$R \leftarrow \mathcal{M}_T(E^T; X), \tag{2}$$

including everything observed by the server. The structural branch is

$$s_R(u, v) = F_1(R, X, u, v; \omega_1). \tag{3}$$

Neither branch may read a private edge, degree, neighborhood, or unprotected embedding at inference. Hence any number of candidate-pair scores is post-processing of R and public inputs.

C. Registered utility

For each certification edge $e = \{u, v\}$, a public seeded map q replaces one endpoint without graph-dependent rejection. False-negative corruptions are part of the estimand. For a scorer s, define

$$U_s(e) = \mathbb{1}\{s(e) > s(q(e))\} + \tfrac{1}{2}\mathbb{1}\{s(e) = s(q(e))\}. \tag{4}$$

The bounded structural advantage is

$$d_R(e) = U_{s_R}(e) - U_{s_0}(e) \in [-1, 1]. \tag{5}$$

The target of the R5 theorem is the mean of d_R over one registered finite holdout population. This pairwise target is related to ranking quality but is not standard ROC-AUC unless the corruption distribution equals the AUC negative distribution.

D. Adversary and released outputs

The primary adversary is an honest-but-curious server that sees every noised client message, the final DP state, the noised certification statistics, the binary decision, and all post-processed scores. Ideal secure aggregation is a secondary trust model in which the server sees only a global aggregate. Secure aggregation changes utility by reducing aggregate noise; it is not itself a DP proof.

The deployed output is $(R, \widetilde{S}, \widetilde{n}, A)$ and scores from the selected branch, where $A \in \{0, 1\}$ is the activation decision. Publicly released research metrics on E^Q are evaluation evidence, not outputs of the deployed DP service.

IV. CERTFED-LP

A. Private training branch

CertFed-LP is architecture-agnostic. An admitted trainer must provide its full RDP curve $\rho_T(\lambda)$ for every registered order $\lambda > 1$ and an inference-closed state R. In R5 we instantiate the candidate with a GAP-style LP adaptation: public descriptors are projected, each client computes bounded neighbor sums, Gaussian noise is applied per hop, and cached normalized channels are scored by mean cosine. We call this a *GAP-style adaptation*, not an official GAP reproduction.

For one hop, changing an undirected edge $\{u, v\}$ changes two bounded node aggregates. The query has ℓ_2 sensitivity $\sqrt{2}$. If each visible client message uses coordinate noise standard deviation σ_T, the aggregate contains standard deviation $\sqrt{K}\sigma_T$, although privacy composes in parallel across unique edge owners. Under ideal secure aggregation, one aggregate noise draw suffices.

B. Private utility release

Condition on any realized training release $R = r$. Certification computes

$$S_r = \sum_{e \in E^C} d_r(e), \qquad n_C = |E^C|. \tag{6}$$

One add/remove certification edge changes (S_r, n_C) by at most $(1, 1)$ in absolute coordinates, so

$$\Delta_2(S_r, n_C) \le \sqrt{2}. \tag{7}$$

Let σ_{abs} denote the absolute coordinate noise standard deviation. The mechanism releases

$$(\widetilde{S}, \widetilde{n}) = (S_r, n_C) + \mathcal{N}(0, \sigma_{\text{abs}}^2 I_2). \tag{8}$$

For visible client messages, independent local noise yields effective aggregate coordinate deviation $\nu = (\sum_k \sigma_{\text{abs},k}^2)^{1/2}$; with equal client noise this is $\sqrt{K}\sigma_{\text{abs}}$. For ideal aggregation, $\nu = \sigma_{\text{abs}}$.

C. One-sided activation certificate

Let $\beta_S, \beta_n, \beta_{\text{stat}}$ be registered failure allocations and

$$b_j = \nu \Phi^{-1}(1 - \beta_j/2), \quad j \in \{S, n\}. \tag{9}$$

Define

$$S_L = \widetilde{S} - b_S, \tag{10}$$
$$n_L = \widetilde{n} - b_n, \qquad n_U = \widetilde{n} + b_n. \tag{11}$$

The mechanism abstains unless $S_L > 0$, $n_L \ge n_{\min}$, and $n_U > 0$. Otherwise the conservative lower certificate is

$$L_{\text{pop}} = \frac{S_L}{n_U} - \sqrt{\frac{2\log(1/\beta_{\text{stat}})}{n_L}}. \tag{12}$$

The deployed action is

$$A = \mathbb{1}\{L_{\text{pop}} \ge \gamma\}. \tag{13}$$

If $A = 1$, the service uses s_R; otherwise it uses s_0. The threshold γ encodes a material gain, not merely non-inferiority.

D. Privacy accounting

A Gaussian query with sensitivity Δ and coordinate standard deviation σ has RDP [17]

$$\rho(\lambda) = \frac{\lambda \Delta^2}{2\sigma^2}. \tag{14}$$

When the edge-role partition is stable, training and conditional certification can use adaptive parallel composition. On its registered role-labelled database, R5 instead takes the conservative sequential sum:

$$\rho_{\text{total}}(\lambda) = \rho_T(\lambda) + \rho_C(\lambda), \tag{15}$$

then converts once,

$$\epsilon(\delta) = \min_{\lambda > 1} \left[\rho_{\text{total}}(\lambda) + \frac{\log(1/\delta)}{\lambda - 1} \right]. \tag{16}$$

Each privacy-grid cell is an alternative deployment. Releasing all cells would require further composition. Sequential summation avoids claiming parallel credit; it does not repair an unstable raw-graph-to-role transformation or extend the registered adjacency relation.

V. Guarantees and Feasibility

This section separates what can be guaranteed from what must be measured. All statements concern the bounded pairwise utility in Section III; none is an AUC confidence interval.

A. Why training-only selection is insufficient

Theorem 1 (Training-only non-identifiability): Fix any training transcript R and public inputs. Over an unrestricted family of target-edge distributions, no selector that observes only (R, X) can be both nonvacuous and guarantee that the selected structural branch has nonnegative target utility relative to the public branch.

Proof sketch. Construct two target worlds with identical training data, public inputs, and therefore identical R, but reverse the probability mass of candidate pairs on which s_R and s_0 disagree. Any training-only selector has the same decision distribution in both worlds, while the utility difference changes sign. A distribution-free no-harm rule must therefore abstain whenever the two scorers can disagree. The complete construction is in Section A.

The theorem does not say that metadata-based selectors are useless on a restricted graph family. It says that their safety is an empirical generalization assumption. CertFed-LP instead obtains direct, private evidence from the target domain.

B. Transcript privacy

Theorem 2 (Adaptive role-wise transcript privacy): Suppose the public role assignment is edge-stable, $\mathcal{M}_T$ is $\rho_T(\lambda)$-RDP on E^T, and, uniformly for every fixed training release r, $\mathcal{M}_C(\cdot; r)$ is $\rho_C(\lambda)$-RDP on E^C. If inference uses only released state and public inputs, the joint transcript is

$$\rho_{\text{joint}}(\lambda) = \max\{\rho_T(\lambda), \rho_C(\lambda)\}\text{-RDP}. \tag{17}$$

On the registered role-labelled database, the conservative sequential bound $\rho_{\text{joint}} \le \rho_T + \rho_C$ is also valid without taking parallel-composition credit. This sum does not by itself establish privacy under adjacency of the raw graph before role assignment.

The proof conditions on the one role-labelled record containing the neighboring edge. Post-processing handles certification after a training-edge change, while uniform conditional DP handles a certification-edge change. R5 reports the sequential sum under its registered role-labelled adjacency because its historical benchmark split was not designed to justify a stronger raw-graph partition theorem.

C. Finite-population no-harm certificate

Let H be the sealed positive holdout of size N and

$$\Delta_H(R) = \frac{1}{N}\sum_{e\in H} d_R(e). \tag{18}$$

An edge-keyed SHA-256 map assigns each $e \in H$ to certification with probability p_C; $Q5$ is the complement. The random-sampling statement below is conditional on the registered random-oracle (or equivalently a secret-key PRF) interpretation of this map. It is not an unconditional information-theoretic claim about deterministic SHA-256 outputs.

Theorem 3 (Finite-holdout no-harm): Conditional on (R, H, n_C), the certification subset is a simple random sample without replacement. With the certificate in Equation (12),

$$\Pr(A = 1 \ \wedge \ \Delta_H(R) < \gamma \mid R, H) \le \beta_S + \beta_n + \beta_{\mathrm{stat}}. \tag{19}$$

The edge-keyed map is edge-stable: adding an edge does not reassign existing edges. Under the stated pseudorandomness assumption, conditional on the realized bounded values and certification count, shared graph endpoints do not alter the random-sampling argument. A one-sided Serfling bound [16] gives the sampling penalty in Equation (12); R5 conservatively sets its finite-population correction to one. The result certifies the registered finite holdout, not future edges or another graph.

D. When private certification is possible

Write $\alpha = \Delta - \gamma > 0$ for the effect above the material threshold, $\chi \ge 1$ for an effective dependence factor, and ν for aggregate Gaussian noise. The sufficient boundary derived in Section D has the principal form

$$n = O\left(\frac{\chi \log(1/\beta)}{\alpha^2} + \frac{\nu\sqrt{\log(1/\beta)}}{\alpha} + n_{\min}\right). \tag{20}$$

Since Gaussian calibration gives $\nu = \widetilde{O}(1/\epsilon)$ under ideal aggregation, the privacy term is $\widetilde{O}(1/(\epsilon\alpha))$. Visible independent client messages replace ν by $\sqrt{K}\nu$ in this implementation; this is a utility cost, not extra privacy composition.

Theorem 4 (Necessary count; informal): For block-replicated Bernoulli hard instances with dependence χ, every central (ϵ, δ)-DP selector with fixed testing error requires, up to constants and approximate-DP logarithms,

$$n = \Omega\left(\frac{\chi}{\alpha^2} + \frac{1}{\epsilon\alpha}\right). \tag{21}$$

Thus the sufficient and necessary boundaries match in their principal dependence on evidence, privacy, and effect size. This does not make the specific Gaussian certificate minimax optimal in δ, nor does it prove that arbitrary graph edges follow the hard-instance model. The lower bound uses the private Le Cam inequality of [15].

VI. Experiments

A. Questions and protocol

We ask four questions: (Q1) Is structural utility heterogeneous enough to make fallback necessary? (Q2) Does CertFed-LP activate nonvacuously while avoiding materially harmful activations? (Q3) How close is the certified policy to an oracle that sees the evaluation sign? (Q4) How do privacy allocation and visibility change the feasible activation region?

TABLE II
Datasets and frozen candidate configuration. Edges are canonical undirected edges.

Dataset	Nodes	Edges	Features	(r, L)
BlogCatalog	10,312	333,983	39	(16,2)
Facebook	22,470	170,823	4,714	(32,1)
PolBlogs	1,490	16,715	2	(8,2)
LastFM Asia	7,624	27,806	7,842	(32,1)
GitHub Social	37,700	289,003	4,005	(32,1)
Deezer Europe	28,281	92,752	31,241	(8,1)

TABLE III
Preregistered primary result over five seeds. "Full" is the finite holdout candidate advantage; "Policy" is disjoint-$Q5$ gain after the private decision.

Dataset	Activate	Full	Policy	AUC pol.
BlogCatalog	5/5	+0.2621	+0.2618	+0.3320
GitHub Social	5/5	+0.1260	+0.1257	+0.1948
PolBlogs	5/5	+0.1254	+0.1251	+0.1757
Deezer Europe	0/5	+0.0337	0.0000	0.0000
Facebook	0/5	−0.0543	0.0000	0.0000
LastFM Asia	0/5	−0.0146	0.0000	0.0000
Mean / total	15/30	+0.0797	**+0.0854**	+0.1171

The confirmatory protocol, configuration hash, code commit, certificate threshold, access gates, and random salts were frozen before the sealed holdout was opened. Six social and blog networks were evaluated with five seeds, producing 30 primary dataset–seed cells. Training, certification, and $Q5$ evaluation roles are disjoint. Hyperparameters were selected only on the earlier validation split. The sealed test payloads were accessed once; all commitments and accountants were independently reproduced.

Here r is projection dimension and L is the number of private aggregation hops. The public branch is sparse cosine over registered node descriptors. The candidate averages cosine scores from the public and cached noisy aggregation channels. It is a GAP-style adaptation, not an official GAP reproduction. Five public home-client partitions are used. The primary mechanism exposes individually noised client messages.

B. Privacy and certificate settings

Training and certification each target $\epsilon = 4$ at $\delta = 10^{-6}$. Their full RDP curves are summed and converted once, yielding $(\epsilon, \delta) = (5.6640, 2\times10^{-6})$. The material threshold is $\gamma = 0.02$, minimum private lower count is 50, and failure allocations are $(\beta_S, \beta_n, \beta_{\mathrm{stat}}) = (0.01, 0.01, 0.03)$. The diagnostic grid contains 5 training budgets, 5 certification budgets, 2 visibility models, 6 datasets, and 5 seeds (1500 records). Every grid cell is a counterfactual deployment, not a jointly released mechanism.

C. Primary result

Figure 2 and Table III answer Q1–Q3. Structural utility changes sign: always-structural is harmful in 10/30 cells.

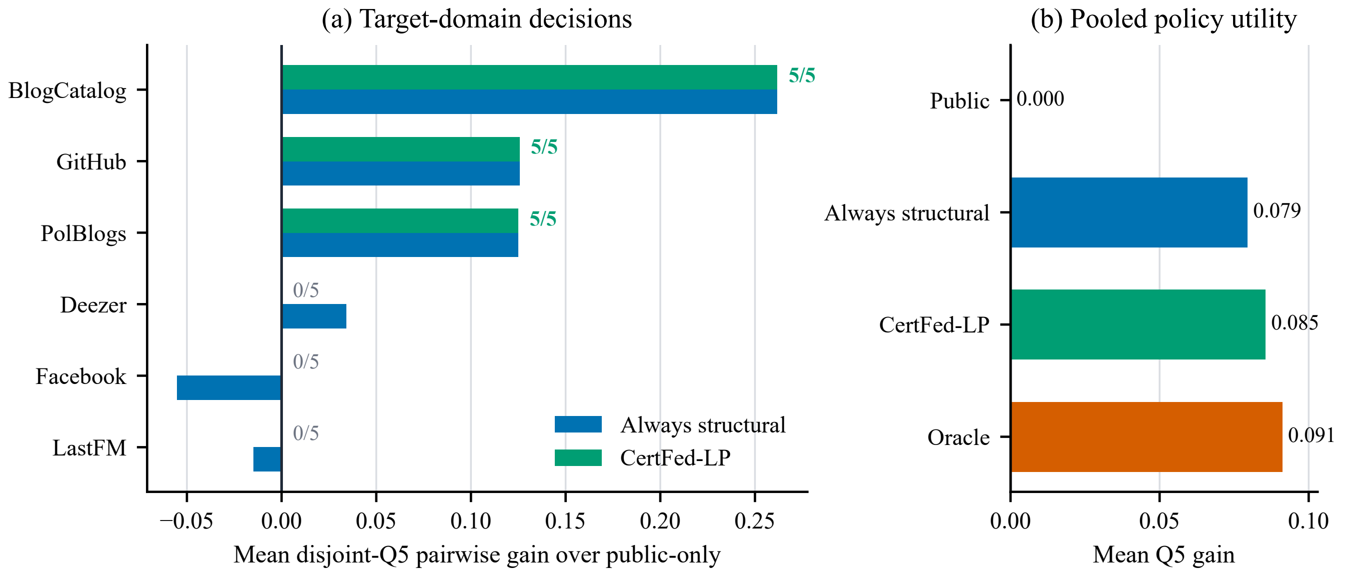

Fig. 2. Primary visible-message result at composed $(\epsilon, \delta) = (5.6640, 2 \times 10^{-6})$. (a) Mean disjoint-$Q5$ pairwise advantage of always enabling the structural branch; green overlays indicate the gain retained by CertFed-LP, with activation counts shown per dataset. (b) Pooled policy gain over public-only. The oracle uses the $Q5$ sign and is not deployable.

CertFed-LP activates exactly all five seeds on BlogCatalog, GitHub Social, and PolBlogs, and falls back on Deezer, Facebook, and LastFM. It observes zero material false activations and has mean $Q5$ policy gain 0.08543. The always-structural and non-deployable oracle gains are 0.07945 and 0.09112, respectively, leaving mean oracle regret 0.00568. As a secondary standard metric, the selected policy improves global ROC-AUC by 0.1171 on average. Deezer is an informative abstention: its mean advantage is positive but the mean certificate lower bound is $-0.0028 < \gamma$.

D. Privacy–utility phase behavior

Figure 3 answers Q4. More training budget expands activation coverage. At the primary $(4, 4)$ target, visible messages activate 15 records with mean policy gain 0.0854, whereas ideal aggregation activates 20 with gain 0.1075. Across all 1500 diagnostic records, no material false activation is observed. This is descriptive evidence beyond the one frozen primary cell; the formal error guarantee remains per deployed certificate.

E. Audit and robustness checks

The independent audit verifies $1500/1500$ unique records, all source and split commitments, the single test-access record, no test tuning, and exact reproduction of every stored RDP curve and converted epsilon. Original global/intra/cross ROC-AUC values are retained only as secondary diagnostics: the certificate and decision use the bounded pairwise estimand. Ties receive half credit, and endpoint corruption performs no graph-dependent rejection, preserving the stated sensitivity.

VII. Discussion and Limitations

a) What the result establishes.: CertFed-LP is a deployment policy, not a new graph encoder. It establishes that a frozen inference-closed structural branch can be privately enabled only where direct target-domain evidence supports a material pairwise gain. The main conclusion is certified fallback under heterogeneous utility, not universal superiority of private graph structure.

b) Finite versus future populations.: The no-harm theorem covers the registered sealed holdout population. An operational system that must generalize to future links needs an additional sampling assumption, temporal certification window, or repeated privacy accounting. Shared nodes do not invalidate the finite random-partition proof, but they can matter for extrapolation beyond that population.

c) Randomization contract.: The R5 partition uses a frozen edge-keyed SHA-256 map and its sampling theorem adopts the registered random-oracle/PRF interpretation. Because the sealed population was fixed before the salt was frozen and opened, it was not chosen adaptively against the realized assignment. A deployment in which data providers can select records after seeing a public salt requires a stronger protocol: for example, a committed secret PRF key hidden from providers until the population is fixed, or an auditable random draw after population freeze. Deterministic SHA-256 alone is not information-theoretic random sampling.

d) Research metrics and deployment outputs.: The DP transcript includes training messages, inference-closed state, noised certificate statistics, the decision, and post-processed score queries. Unprotected embeddings, graph-backed inference, public test metrics, and simultaneous release of the diagnostic grid are outside the guarantee.

e) Candidate dependence.: Nonvacuity depends on the admitted learner. A stronger candidate can increase the number of safe activations; a weak one correctly produces fallback. Our GAP-style candidate is deliberately simple and is not claimed to dominate existing private graph learners. Certifying learned

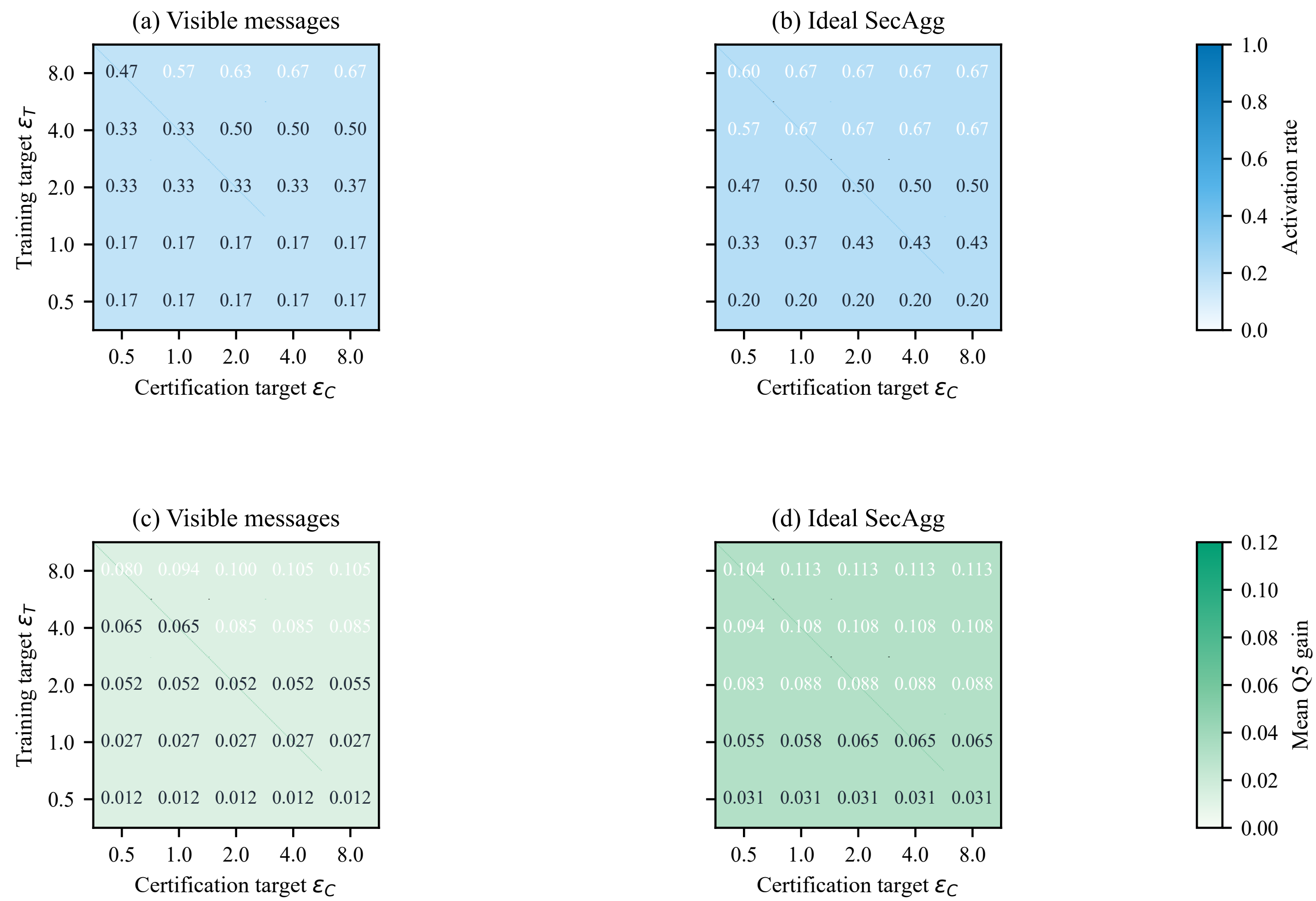

Fig. 3. Diagnostic activation phase over training budget, certification budget, and visibility. Each cell reports a counterfactual deployment over 30 dataset–seed records. Ideal secure aggregation is a separate trust model and reduces aggregate noise; it is not an extra DP claim.

neural scorers and temporal target populations is a direct next step.

f) Trust and client noise.: Visible independent messages incur aggregate standard deviation proportional to $\sqrt{K}$ in this implementation. Ideal secure aggregation removes that utility penalty but assumes an additional trusted functionality. DP alone does not hide individual messages from the server.

g) Statistical multiplicity.: The confirmatory claim is tied to one preregistered primary cell. The phase diagram is diagnostic. Deploying multiple candidate branches, thresholds, or privacy cells on one graph requires privacy composition and statistical multiplicity control.

VIII. Conclusion

Federated edge-private link prediction does not reduce to choosing a graph encoder with high average accuracy. A private structural branch can help one target graph and harm another, while training-only evidence cannot provide a distribution-free nontrivial no-harm decision. CertFed-LP turns this heterogeneity into an explicit deployment problem: train an inference-closed edge-DP candidate, privately certify its bounded target-domain advantage, and fall back to a public-only scorer when evidence is insufficient. Its transcript privacy, finite-population safety guarantee, and near-matching feasibility bounds state exactly what is protected and certified. The preregistered six-network study shows that the policy is both conservative and nonvacuous. These results position private utility certification, rather than unconditional structural deployment, as a practical foundation for differentially private link prediction in federated settings.

Appendix A
Proof of Training-Only Non-Identifiability

Let T contain every quantity available before target-domain certification, including public descriptors, the private training transcript, privacy parameters, and arbitrary training-derived graph statistics. Fix a realization $T = t$ and two nonempty sets of future comparison records $\mathcal{A}_+$ and $\mathcal{A}_-$ that are not inputs to T. Choose the fixed scorers so their bounded utility difference is $+a$ on $\mathcal{A}_+$ and $-a$ on $\mathcal{A}_-$ for some $a > 0$. Let P_+ and P_- place target mass on $\mathcal{A}_+$ and $\mathcal{A}_-$, respectively, while keeping all inputs to T identical.

If a randomized T-measurable selector activates with probability $p(t)$, its expected gain is $p(t)a$ under P_+ and $-p(t)a$ under P_-. A no-harm guarantee under P_- forces $p(t) = 0$, contradicting nonvacuous activation under P_+. Moreover, relative to a sign oracle, the regrets are $(1-p)a$ and pa; minimax regret is at least $a/2$. Target-domain certification escapes the construction because its input distributions differ in the two worlds.

Appendix B
Proof of Transcript Privacy

A canonical neighboring edge has one public role. If it belongs to E^T, E^C is identical in the neighboring databases. Conditional certification is the same Markov kernel applied to the two possible training releases, so appending it and the decision is post-processing of $\mathcal{M}_T$. If the edge belongs to E^C, the distribution of R is identical. Uniform conditional RDP of $\mathcal{M}_C(\cdot; r)$ remains bounded by $\rho_C(\lambda)$ after integrating over that common distribution. An E^Q edge affects no deployed mechanism. Taking the worse of the two affected cases proves Theorem 2.

For the certification query $Q_C = (S, n)$, adding or removing one edge changes one bounded contribution by at most one and the count by one, hence

$$\Delta_2 Q_C \leq \sqrt{1^2 + 1^2} = \sqrt{2}. \tag{22}$$

With absolute coordinate noise σ_{abs}, its Gaussian RDP is

$$\rho_C(\lambda) = \frac{\lambda \Delta_2^2}{2\sigma_{\text{abs}}^2}. \tag{23}$$

The (ϵ, δ) conversion used in every result record is

$$\epsilon(\delta) = \min_{\lambda \in \Lambda} \left\{ \rho_{\text{joint}}(\lambda) + \frac{\log(1/\delta)}{\lambda - 1} \right\}. \tag{24}$$

R5 sums the stored training and certification RDP arrays order by order and then performs this conversion once.

Appendix C
Proof of the Finite-Holdout Certificate

Condition on (R, H, n_C). Under the registered random-oracle/PRF interpretation of the edge-keyed SHA-256 map, its outputs are independent across the fixed keys, so conditioning on the count makes E^C a uniform size-n_C subset of H. This is a computational-randomization assumption, not a theorem about deterministic SHA-256. The values $d_R(e) \in [-1, 1]$ are then a fixed finite population. For $\widehat{\Delta}_C = n_C^{-1} \sum_{e \in E^C} d_R(e)$, one-sided sampling-without-replacement concentration gives

$$\Pr\{\widehat{\Delta}_C - \Delta_H \geq t \mid R, H, n_C\} \leq \exp\left(-\frac{n_C t^2}{2 f_N}\right), \tag{25}$$

where $f_N = 1 - (n_C - 1)/N \leq 1$. R5 uses $f_N = 1$ in the decision.

Let $\mathcal{G}_S$ and $\mathcal{G}_n$ be the Gaussian events on which the released sum and count deviate by no more than b_S and b_n. On their intersection, $S_L \leq S$, $n_L \leq n_C \leq n_U$, and, when $S_L > 0$,

$$\frac{S_L}{n_U} \leq \widehat{\Delta}_C. \tag{26}$$

The sampling event implies

$$\Delta_H \geq \widehat{\Delta}_C - \sqrt{\frac{2\log(1/\beta_{\text{stat}})}{n_C}} \geq L_{\text{pop}}. \tag{27}$$

A union bound over the two Gaussian events and the sampling event proves Theorem 3. Invalid counts or a nonpositive corrected numerator cause abstention and cannot create a false activation.

Appendix D
Sufficient Feasibility Boundary

Assume bounded utilities with mean Δ satisfy

$$\Pr\{\overline{U} \leq \Delta - t\} \leq \exp\left(-\frac{nt^2}{2\chi}\right). \tag{28}$$

For an additional power failure β_p, define

$$a_p(n) = \sqrt{\frac{2\chi \log(1/\beta_p)}{n}}, \tag{29}$$

$$a_v(n) = \sqrt{\frac{2\chi \log(1/\beta_{\text{stat}})}{n}}. \tag{30}$$

On the sampling and Gaussian good events, the operational certificate is at least

$$B(n, \Delta) = \frac{n[\Delta - a_p(n)] - 2b_S}{n + 2b_n} - a_v(n - 2b_n). \tag{31}$$

This expression is valid when $n - 2b_n \geq n_{\min}$ and the numerator is positive. Consequently, $B(n, \Delta) \geq \gamma$ is sufficient for activation with the registered high probability. Solving for Δ yields

$$\Delta_{\min}(n) = a_p(n) + \frac{2b_S}{n} \tag{32}$$

$$+ \frac{n + 2b_n}{n} \left[\gamma + a_v(n - 2b_n)\right]. \tag{33}$$

With no privacy noise, this reduces to $\gamma + a_p(n) + a_v(n)$.

Appendix E
Necessary Count

For completeness, consider the registered block-replicated Rademacher hard pair. Under H_0 each independent block is positive with probability $p_0 = (1+\gamma)/2$; under H_1 it is positive with probability $p_1 = (1+\gamma+\alpha)/2$. There are n/χ independent blocks and each is replicated χ times. Classical Le Cam and Pinsker inequalities imply

$$n \geq \frac{2\chi(1 - 2\eta)^2}{D_{\text{KL}}(\text{Ber}(p_0) \| \text{Ber}(p_1))}. \tag{34}$$

For the privacy term, convert add/remove (ϵ, δ)-DP to replacement privacy with $\epsilon_r = 2\epsilon$ and $\delta_r = (1 + e^{\epsilon})\delta$. Let $D_\star$ be the smallest nonnegative solution of

$$0.9 e^{-10\epsilon_r D} - 10 D \delta_r \leq 2\eta. \tag{35}$$

Under maximal coupling the expected replacement Hamming distance is $n\alpha/2$. The approximate-DP Le Cam bound in [15] therefore requires

$$n \geq \frac{2D_\star}{\alpha}. \tag{36}$$

For pure DP, this gives $n \geq \log(0.9/(2\eta))/(10\epsilon\alpha)$. Since

$$D_{\text{KL}}(\text{Ber}(p_0) \| \text{Ber}(p_1)) = \frac{\alpha^2}{2(1 - \gamma^2)} + O(\alpha^3), \tag{37}$$

the principal necessary rate is $\Omega(\chi/\alpha^2 + 1/(\epsilon\alpha))$.

APPENDIX F
REPRODUCIBILITY AND CLAIM BOUNDARIES

The accompanying artifact contains the frozen primary configuration, immutable raw outcomes, summary, authoritative one-time access record, and independent audit. The runner is fixed at commit `dcdbf2e`. Every result stores its full RDP array and distinguishes the confirmatory cell from the diagnostic grid.

The following are intentionally not claimed: a confidence interval for ROC-AUC, temporal or cross-domain no-harm, official reproduction or universal superiority of GAP, simultaneous private release of the phase grid, or privacy for an inference path that rereads an unreleased private graph.

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